

Inferred global dense residue graphs from primary structure sequences enable protein interaction prediction via directed graph convolutional neural networks

Islam Akef Ebeid^{1,*}, Haoteng Tang² and Pengfei Gu²

¹Texas Woman's University, The Division of Computer Science, Denton, TX, USA

²The University of Texas Rio Grande Valley, The Department of Computer Science, Edinburg, TX, USA

Correspondence*:

Islam Akef Ebeid
iebeid@twu.edu

2 ABSTRACT

3 Accurate prediction of protein-protein interactions (PPIs) is crucial to understanding cellular
4 functions and advancing drug discovery. While some in-silico methods leverage direct sequence
5 embeddings from Protein Language Models (PLMs), others apply Graph Neural Networks
6 (GNNs) to topological features from PPI networks or 3D protein structures. Here, we introduce
7 a novel approach that models protein primary structure through a hierarchy of globally inferred
8 n-gram graphs. For each n-gram level, residue transition probabilities, aggregated from a large
9 sequence corpus, define the edge weights of an underlying directed graph. We propose a custom
10 directed graph convolutional network, *ProtGramDirectGCN*, featuring a unique convolutional
11 layer that processes information through separate path-specific (incoming, outgoing, undirected,
12 homophilic and heterophilic) and shared transformations. These components are then combined
13 via a learnable gating mechanism. The efficacy of *ProtGramDirectGCN* is first established
14 on standard node classification benchmarks, where its performance is on par with established
15 methods on general benchmark datasets yet highlights its specialization for complex, directed,
16 limited and dense heterophilic graph structures. Subsequently, it is applied to the hierarchy of
17 n-gram graphs to learn residue embeddings, which are then pooled to generate protein-level
18 representations. The main focus of this study is to find less intensive alternative approaches
19 to PLMs for the downstream task of protein interaction link prediction, our method achieves
20 robust predictive power, suggesting that our graph-based representation of global sequence
21 statistics offers a potent and computationally distinct alternative to resource-intensive PLMs on
22 the task of protein protein interaction prediction. Future work includes testing our robust two-stage
23 graph representation learning strategy that effectively captures global sequence statistics and
24 transitional patterns on a wide range of bioinformatics tasks.

25 **Keywords:** uniprot, graph theory, graph representation learning, graph neural networks, graph convolution networks, link prediction,
26 proteomics, protein-protein interaction prediction, large language models, sequence-to-sequence modeling

1 INTRODUCTION

27 Protein-protein interactions form a network of physical contacts and functional associations mediated by
28 molecular bonds. These interactions serve as the basis for cellular processes and are called the cellular

interactome. Therefore, understanding the underlying mechanisms by predicting valid interactions between proteins is the foundation for many *in-vitro* biomedical endeavors, such as understanding disease mechanisms, drug development, repurposing, and the potential development of futuristic biotechnologies Vidal et al. (2011) Scott et al. (2016). *In-vitro* protein interaction prediction methods, including Yeast-2-Hybrid screening, co-immunoprecipitation followed by mass spectrometry and affinity purification, have been used to infer evidence of protein association. However, these methods are usually prone to a high rate of false positives and false negatives Rao et al. (2014).

Computational methods, also known as *in-silico* and data-driven approaches in protein interaction prediction, are crucial in life sciences, as they help alleviate several of the significant challenges of the *in-vitro* methods mentioned above. The initial stages of drug discovery are heavily based on identifying and confirming valid drug targets, often proteins. This activity is typically protracted, resource-intensive, and time-consuming. Subsequent *in-vitro* screening of drug candidates against potential targets cannot be done efficiently until a validated list of candidate proteins is established. Delays in this upstream target identification task lead to delays in the beginning of extensive *in-vitro* studies Scannell et al. (2012). Therefore, *in-silico* methods, mainly relying on the predictive power of complex machine learning models, are not meant to replace *in-vitro* methods. Instead, they are integrated into the workflow to create a potential pool of valid interactions waiting for wet lab filtering and confirmation.

Recent advancements in machine learning, neural networks and deep learning approaches that automate feature extraction and embed biological entities into a real vector space, where meaningful algebraic operations can be performed on the learned vectors representing individual residues or proteins. The advent of the transformer architecture as sequence to sequence architectures such as recurrent neural networks based networks like long-short term memory (LSTM) have shown a lack of understanding of context incorporation in language modeling has revived the sought after context aware networks. Thought that contextual understanding have had glimpses in non recurrent language based neural networks like word2vec where the goal becomes incorporating context via binary negative loss function that classifies in and out of context window words to the current word. The introduction of the attention mechanism and the transformer architecture however has increased the predictive power of language models by orders of magnitudes on multiple tasks which has contributed to the proliferation of different designs and architectures like BERT, T5 and GPT Vaswani et al. (2017). That however happens at a great computational, and environmental cost via the increase data hunger for these architectures, in addition to the near linear correlation between model's predictive power and the number of parameters present in the network. Further more these architectures have been fine tuned on specific biomedical applications; for example BioBERT fine tunes BERT over the corpus of PubMed metadata and available publications on multiple tasks one of which is biological named entity recognition and extraction for names of diseases, genes, proteins, species and drugs. The protein embeddings extracted from BioBERT could provide contextual meaning in downstream tasks of protein interaction prediction or gene identification.

In drug development especially in protein protein interaction prediction (PPI) advanced models have been applied on multiple levels. For example in reinforcement learning a prominent recent advancement is Google's AlphaFold Jumper et al. (2021). This complex model aims to predict protein three-dimensional structures from primary sequences, a central challenge in biomedical informatics known as protein folding prediction. Predicted three-dimensional structures are often utilized in frameworks that aim to predict protein interactions from all levels of protein structure representation via combining features from the first, second, and third levels and structural features from protein-protein interaction networks Zhou et al. (2022) Jha et al. (2022). However, the most common approaches in *in-silico* protein-protein interaction

prediction are primary structure sequence-based methods, where the sequential one-dimensional nature of amino acid and residue-level representations lend themselves to modern language modeling, where the core lies in the context encoded in the transition probabilities between residues. Protein sequences can be conceptualized as a sequence of amino acids (or peptides), analogous to sentences being sequences of words. This analogy allows for the application of large language modeling techniques. For instance, ProtBert applies the BERT architecture to primary protein structures which has yielded accurate models on downstream tasks, generating protein level vector representations. Elnaggar et al. (2022) at the residue and protein levels. Moreover work like ? Before the advent of the transformer architecture, LSTM was the primary architecture used in sequence modeling Cho et al. (2014) Hochreiter and Schmidhuber (1997) for proteins again following the obvious analogy between primary structure sequences and language. The input protein sequence is usually tokenized at different levels or granularities, such as representing single amino acids as words or a group of amino acids as k-mers . The outputs of all of these deep learning architectures are typically pooled to produce per-protein embeddings that capture sequential features Elnaggar et al. (2022).

However, even the most advanced modern approaches have several problems.

- The limited context window size must be larger to capture longer-range dependencies beyond the immediate neighborhood.
- Increasing the number of layers can lead to a significant (potentially exponential) increase in the number of parameters, demanding more computational resources.
- Different granularity of tokenization when applying large language models to protein sequences might be meaningless, especially if it is based on anything other than residues.

Here, we propose a novel approach to modeling protein sequences that partially overcomes some of these limitations. We cast the sequences as random walks sampled according to transition probabilities within a directed n-gram graph G_n of amino acids. The directed graph G_n is inferred from a database of curated protein sequences (UniProt) The UniProt Consortium (2023). Then, a custom-directed graph convolution neural network, *ProtGramDirectGCN*, learns the dense relationships between the n-grams. The learnt representations are then evaluated on a PPI-based link prediction task and compared with other models to establish the method's validity. This approach does not require a context window like in protein large language models Elnaggar et al. (2022). The computational limitations that contribute to limited context windows are only applied to a graph of n-grams, where the first-order neighborhood of a graph convolution operator approximation Kipf and Welling (2017) has a limited effect on the output compared to the sizes of a well-capturing context window in a large language model. This bottom-up approach ensures that the limitations are only applied to the lowest level of representation, overcoming the need for intensive computational power. In addition, tokenization in proteins differs from language, where each word can be treated as a separate entity with its meaning; it is more akin to proteins being represented via sequences of amino acids, and any other grouping is merely for convenience, not inherent meaning Ofer et al. (2021).

The specific transition sequence of amino acids via their side chains or R groups determines how a polypeptide chain will fold. Hydrogen bonds, ionic bonds, and hydrophobic interactions generally drive the folding. And in the process, the local secondary structure, including alpha and beta helices, eventually creates binding sites essential for forming subunit proteins or interacting with other molecules. Hence, our intuition is that the primary structure sequencing and the transition frequencies between residues inform down the line not only the 3-dimensional tertiary structure of the protein but also tell the possible

interactions with other proteins or the quaternary structure Dill and MacCallum (2012) Perkins et al. (2010) Anfinsen (1973).

Accordingly we hypothesize that the global directed dense graph of n-grams G_n encodes the potential relationships between proteins, and that learning accurate vector representations of G_n not only provides comparable performance to PLMs in the task of PPI based link prediction but also offers a method to generate protein embeddings on the fly without the need to store per-protein embeddings nor to fine tune hefty pretrained models. Here we are trying to answer the following research questions:

- Does learning representations of proteins from the embedded, inferred directed graph of n-grams G_n encode valid associations between proteins?
- Is our *ProtGramDirectGCN* model credible and valid?
- What is the predictive power of our hierarchical feature based n-gram representation G_n ?
- Is the performance of the *ProtGramDirectGCN* model comparable to LLMs? And what are the implications of that?

2 METHODS

This section details the design, development, and evaluation of our novel *ProtGramDirectGCN* model, a directed graph convolutional network tailored for learning representations from dense, directed, and weighted graphs. The primary motivation for this model arises from the need to effectively process a global graph of n-grams, G_n , constructed from large-scale protein sequence data where capturing directionality and transition frequencies is paramount for deriving meaningful biological insights.

2.1 Preliminaries

Our approach treats individual proteins, identified via the comprehensive databases like UniProt The UniProt Consortium (2023), as distinct entities. These proteins form the nodes V in a high-level biological graph $G_{PPI} = (V, E_{PPI})$, where edges E_{PPI} represent observed or potential interactions. The main objective is to solve the link prediction problem within G_{PPI} . Which can be formulated such that the probability of observing a link given other links indicating a highly probable interaction is described by $P(e_{new}|E)$ where $e \in E$ and $e_{new} \notin E$ and is usually estimated by the formula $\frac{4m}{(n^2)(n-1)^2}$ where the probability and is significantly less probable than observing an existing link where the probability of observing an existing link is $\frac{2m}{n(n-1)}$ which demonstrates the inherent difficulty of the PPI task. A foundational aspect of our methodology is the detailed representation of individual protein sequences through a hierarchy of n-gram graphs. For a given residue n , each protein P_i is defined by its primary amino acid sequence $R_i = (r_1, r_2, \dots, r_L)$, which can be decomposed into a sequence of overlapping n-grams following the general Markov property assumption with probabilities: $P(n_i|n_{i-(N-1)}, \dots, t_{i-1}) = \frac{C(n_{i-(N-1)}, \dots, t_{i-1}, t_i)}{C(n_{i-(N-1)}, \dots, t_{i-1})}$ $P(T) \approx \prod_{i=1}^N P(n_i|t_{i-(N-1)}, \dots, t_{i-1})$ Where n is a single gram where one gram is one token one amino acid symbol. Two grams is 2 amino acid symbols in the window and so on and N is the vocab size. We conceptualize these sequences through a global directed, dense graph of n-grams, $G_n = (V_n, E_n)$. V_n is the finite set of unique n-gram types observed in a large corpus. E_n represents directed transitions between n-gram types. Hence an edge (u, v) from n-gram u to v exists if v can be formed by shifting a one-character window over u . For example, for $n = 3$, an edge exists from 'ACG' to 'CGT'. Each edge is assigned a weight w_{uv} , corresponding to the observed frequency of this transition across the corpus. A valid protein sequence $R = (r_1, r_2, \dots, r_L)$ is thus viewed as a specific path or

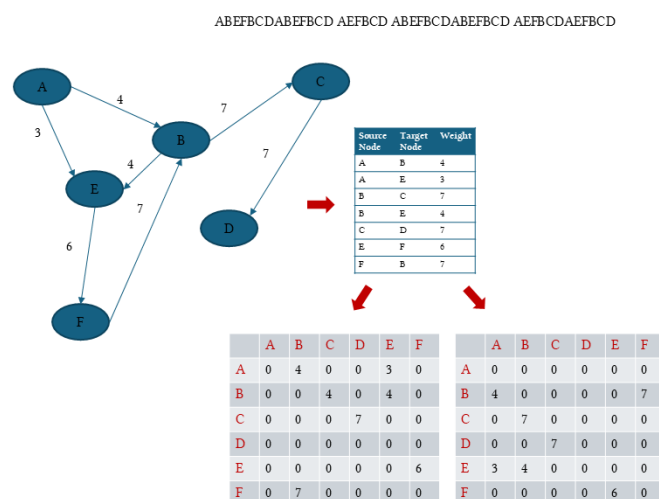


Figure 1. An example of dummy sequences separated by a space indicating multiple proteins. The figure shows how the transitions between the residues or characters are conceptualized as a directed, weighted, dense graph where the weights are the transition frequencies calculated as counts or probabilities. In addition, the figure shows how we split the directed adjacency into an A_{in} and an A_{out} . Later, we describe how to overcome the non-hermitian nature of these two matrices to make them suitable for neural networks.

random walk of length $L - 1$ on G_n on this hierarchy of n-gram graphs. This conceptualization aligns with the idea that protein sequences can be seen as generated from a 'source graph' of amino acid symbols via a random walk process.

Assuming a first-order Markov process, the probability of observing a particular sequence R , given its starting residue r_1 and the transition probabilities derived from G_n , can be formulated. If $P(r_{j+1}|r_j) = \frac{w_{r_j r_{j+1}}}{\sum_k w_{r_j k}}$ is the transition probability from residue r_j to r_{j+1} given normalized edge weights, then the probability of the sequence R is $P(R|r_1, G_n) = \prod_{j=1}^{L-1} P(r_{j+1}|r_j)$. This probabilistic view, rooted in the empirically derived G_n , allows for a nuanced understanding of sequence validity, likelihood, and structure. We aim not to view the amino acid sequence representation as a mere Markovian sequence but also to consider the existence of different relationships between a residue and many other residues. The directed nature of G_n is crucial, naturally modeling the $N - to - C$ terminus directionality of polypeptide chains and the inherent asymmetry of residue relationships. See figure 1. Our custom Directed Graph Convolutional Network (*ProtGramDirectGCN*) is specifically designed to learn from these G_n graphs. Graph Neural Networks (GNNs) are architectures adept at learning node representations by iteratively aggregating information from neighborhoods, also known as message passing Scarselli et al. (2009). GNNs can be applied to the two different graph domains in the first is spatial and the second is spectral. The spatial models perform the message passing via direct computation while the spectral methods rely on operating on the adjacency matrix directly to approximate the convolutional operation. A foundational GNN is the Graph Convolutional Network (GCN) ?, whose layer typically updates node features according to $H^{(l+1)} = \sigma(\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} H^{(l)} W^{(l)})$. Here, $H^{(0)}$ would be initial features for n-gram types in V_n , $\tilde{A}$ is the adjacency matrix of G_n with added self-loops, $\tilde{D}$ is its corresponding diagonal degree matrix for normalization, $W^{(l)}$ is a trainable weight matrix, and σ is a non-linear activation function.

Standard GCNs are primarily for undirected graphs. However multiple works have explored applying GCNs to directed graphs for a complete review of directed GCN methods please see the supplementary material. Our *ProtGramDirectGCN* adapts this framework to effectively learn from the directed and

weighted edges of G_n , separating information flow based on edge existence, directionality and the homophily property. Our model learns rich embeddings $V_{n,k} \in \mathbb{R}^{d'}$ for each n-gram type k . Rich feature vectors for entire protein sequences P_i are generated by aggregating their constituent n-grams' learned *ProtGramDirectGCN* embeddings. These protein-level feature vectors are utilized in the downstream link prediction task within G_{PPI} . Next we will describe the architecture of the *ProtGramDirectGCN* model.

2.2 ProtGramDirectGCN

2.2.1 Propagation Matrix Formulation

The graph structure is initially captured by a raw weighted adjacency matrix $A_{raw} \in \mathbb{R}^{N \times N}$, where $(A_{raw})_{uv} = w_{uv}$. From this, we define the out-degree weighted adjacency matrix $A_{out}^{(w)} = A_{raw}$ and the in-degree weighted adjacency matrix $A_{in}^{(w)} = A_{raw}^T$. In addition we also generate the structural symmetric undirected adjacency A . And more importantly we generated the homophily and heterophily path specific undirected adjacencies A_{homo} and A_{hetero} . A key component of *ProtGramDirectGCN* is a specific preprocessing step for these adjacency matrices, designed to create stable and informative propagation matrices. For a given weighted adjacency matrix $A^{(w)}$ (either $A_{out}^{(w)}$ or $A_{in}^{(w)}$), we first compute its row-normalized counterpart $A^{(n)} = D^{-1}A^{(w)}$, where D is the diagonal out-degree matrix. To overcome the non-hermitian nature of $A^{(n)}$, we compute its symmetric-like (S) and skew-symmetric-like (K) components:

$$S = \frac{A^{(n)} + (A^{(n)})^T}{2} \quad \text{and} \quad K = \frac{A^{(n)} - (A^{(n)})^T}{2} \quad (1)$$

The final propagation matrix $\mathcal{A}$ is derived from the element-wise magnitude of these components, with an added identity matrix I for self-loops:

$$\mathcal{A} = \sqrt{S^2 + K^2 + \epsilon} + I \quad (2)$$

where the square operations are element-wise and ϵ is a small constant (e.g., 1×10^{-9}) for numerical stability. This process yields two distinct propagation matrices, $\mathcal{A}_{out}$ and $\mathcal{A}_{in}$, which are used for message aggregation in the convolutional layers. This construction aims to capture both the symmetric and anti-symmetric aspects of the directed relationships, offering a more robust representation of directed influence in addition to the other paths $\mathcal{A}$ and $\mathcal{A}_{\square\square\square}$ and $\mathcal{A}_{\langle\rangle\rangle}$.

2.2.2 Propagation Layer

Given node features $H^{(l)} \in \mathbb{R}^{N \times F^{(l)}}$ at layer l , the layer computes the features for the next layer by processing information through five distinct channels: incoming, outgoing, homophilic association, heterophilic association and undirected. Each channel combines a standard GCN message passing operation with a feed forward layer as final feature transformation. For the incoming path, the aggregated message is a combination of a propagated component and a shared feature transformation:

$$H_{in}^{(l+1)} = \left(\mathcal{A}_{in}(H^{(l)}W_{main,in}^{(l)} + b_{main,in}^{(l)}) \right) + \left(H^{(l)}W_{shared}^{(l)} + b_{shared,in}^{(l)} \right) \quad (3)$$

Similarly, for the outgoing path:

$$H_{out}^{(l+1)} = \left(\mathcal{A}_{out}(H^{(l)}W_{main,out}^{(l)} + b_{main,out}^{(l)}) \right) + \left(H^{(l)}W_{shared}^{(l)} + b_{shared,out}^{(l)} \right) \quad (4)$$

211 Similarly, for the homophilic association path:

$$H_{homo}^{(l+1)} = \left(\mathcal{A}_{homo}(H^{(l)}W_{main,homo}^{(l)} + b_{main,homo}^{(l)}) + \left(H^{(l)}W_{shared}^{(l)} + b_{shared,homo}^{(l)} \right) \right) \quad (5)$$

212 Similarly, for the heterophilic association path:

$$H_{hetero}^{(l+1)} = \left(\mathcal{A}_{hetero}(H^{(l)}W_{main,hetero}^{(l)} + b_{main,hetero}^{(l)}) + \left(H^{(l)}W_{shared}^{(l)} + b_{shared,hetero}^{(l)} \right) \right) \quad (6)$$

213 And for the undirected path, using a standard symmetrically normalized adjacency matrix $\tilde{A}_{undir}$:

$$H_{undir}^{(l+1)} = \left(\tilde{A}_{undir}(H^{(l)}W_{main,undir}^{(l)} + b_{main,undir}^{(l)}) + \left(H^{(l)}W_{shared}^{(l)} + b_{shared,undir}^{(l)} \right) \right) \quad (7)$$

214 In addition we model the idea of positional encoding which ensures that the model has some notion of time
215 and sequence. We do that by adding a non transformed learnable embedding layer that gives each node
216 (n-gram) its positional identity:

$$B_{const}^{(l)} \in \mathbb{R}^{n \times d} \quad (8)$$

217 where $W_{main,*}$ are path-specific weight matrices and W_{shared} is a single weight matrix shared across all
218 three paths, acting on the original node features. In addition d is a chosen dimension. These five processed
219 signals are then combined using a learnable, node-wise gating mechanism alongside a separate learnable
220 feature vector that captures the node positional identity in the graph. Eventually the model resembles an
221 algebraic multivariate first order polynomial linear combination of features that represent separate yet
222 integrated graph properties $aX + bY + cZ + d$:

$$\begin{aligned} H_{pre-activation}^{(l+1)} = & \left(C_{undir}^{(l)} \odot H_{undir}^{(l+1)} \right) + \left(C_{in}^{(l)} \odot H_{in}^{(l+1)} \right) \\ & + \left(C_{out}^{(l)} \odot H_{out}^{(l+1)} \right) + \left(C_{homo}^{(l)} \odot H_{homo}^{(l+1)} \right) \\ & + \left(C_{hetero}^{(l)} \odot H_{hetero}^{(l+1)} \right) + B_{const}^{(l)} \end{aligned} \quad (9)$$

223 where $C_*^{(l)}$ are learnable gating vectors that facilitate understanding the importance of the contribution of
224 each path in the learning. Finally, a residual connection is added before applying a Leaky ReLU activation
225 function:

$$H^{(l+1)} = \sigma_{LReLU} \left(H_{pre-activation}^{(l+1)} + H^{(l)}W_{res}^{(l)} \right) \quad (10)$$

226 where $W_{res}^{(l)}$ is a linear projection for the residual connection if feature dimensions change.

227 2.2.3 Model Architecture

228 The full *ProtGramDirectGCN* model is composed of a stack of these custom hybrid layers. The
229 overall architecture is defined as follows:

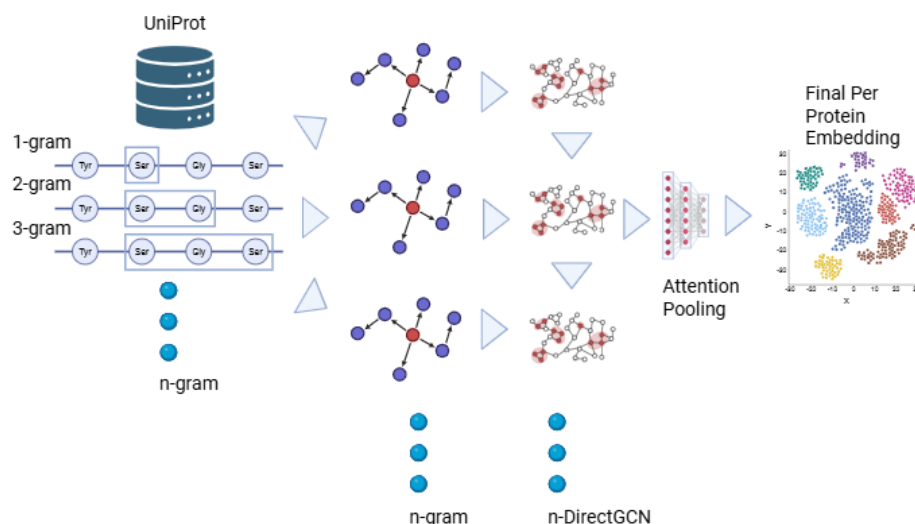
- 230 • **Input Layer:** The initial node features for the n-grams, $H^{(0)} \in \mathbb{R}^{N \times F^{(0)}}$, are either identity initialized
231 (for $n = 1$) or derived from the embeddings of the previous n-gram level (for $n > 1$).
- 232 • **Hidden Layers:** The model stacks L hidden layers. For each layer $l \in \{0, \dots, L - 1\}$, the output
233 $H^{(l+1)}$ is computed by applying the *DirectGCN* layer transformation (Equations 3-7) to the previous
234 layer's output $H^{(l)}$. A residual connection is included to facilitate deeper architectures, and a Leaky

235 ReLU activation function followed by dropout is applied after each layer to introduce non-linearity
236 and prevent overfitting.

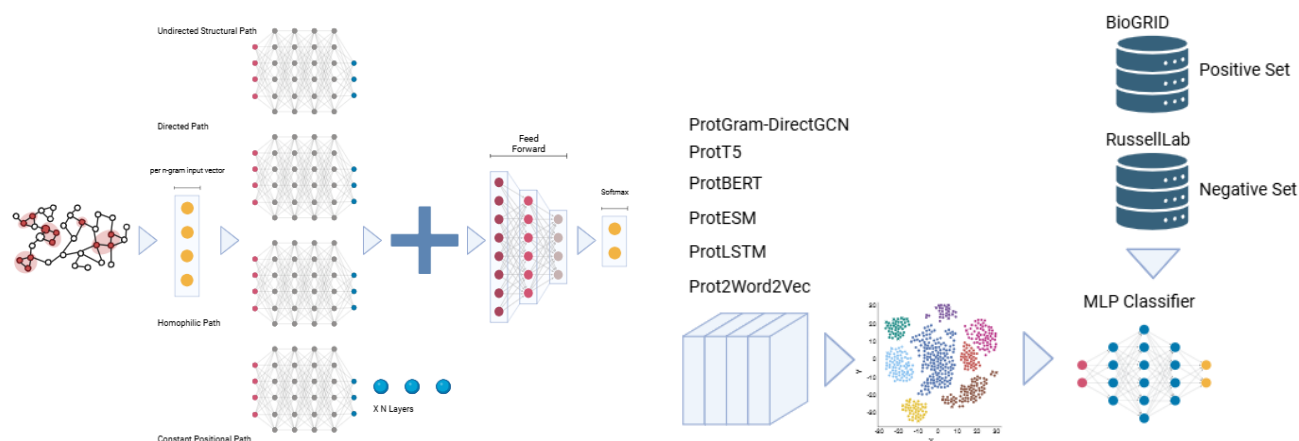
237 • **Output Layer:** The output of the final hidden layer, $H^{(L)}$, serves as the learned n-gram embedding,
238 Z_{n-gram} . For the auxiliary node classification tasks (community detection, next node prediction or
239 masked node prediction) on G_n , these embeddings are passed through a final linear decoder, which is
240 a small feed forward layer, to produce the final class logits:

$$\text{Logits} = \text{Decoder}(Z_{n-gram}) \quad (11)$$

241 A *LogSoftmax* function is then applied to these logits for training with a negative log-likelihood loss.
242 The final embeddings, Z_{n-gram} , are L2-normalized before extraction.



(a) *ProtGram*: our unique approach in modeling protein sequences.



(b) *DirectGCN*: the layer is composite of multiple paths extracted from the graph; unstructured, directional, homophily in addition to a positional identity the paths are then aggregated and transformed via a final feed forward layer and is trained on a next node prediction classification task.

(c) *PPI*: the final embedding are pooled via an attention pooling layer the passed down to the downstream task of classification based link prediction where a classifier is trained on a standard ground truth data.

Figure 2. *ProtGram – DirectGCN*

Note: The proposed architecture; for each of the 5 information paths (Incoming, Outgoing, Undirected, Homophilic, and Heterophilic), input features ($H^{(l)}$) are processed through two parallel paths: a path-specific message passing transformation $W_{main,in}$ and a shared transformation W_{shared} . The results are aggregated and then combined via a learnable gating mechanism controlled by coefficients (C) before a final residual connection and activation.

3 EXPERIMENTS

This section outlines the experimental design employed to evaluate the proposed *ProtGram–DirectGCN* model. Our experiments are structured to: (1) assess the intrinsic performance of *DirectGCN* on standard graph benchmark datasets; (2) detail the construction of a hierarchy of global n-gram graphs G_n from the UniProt sequence dataset a method we call *ProtGram*; (3) evaluate the ability of *ProtGram –*

DirectGCN to learn meaningful representations from these G_n via a self-supervised pre-training task; (4) apply these learned representations, after pooling to the protein level, to the downstream task of *PPI* link prediction; and (5) compare the efficacy of our protein embeddings against those derived from the state-of-the-art PLMs and other baselines.

3.1 Materials

Our approach in this investigation is to minimize any cleaning or modification of publicly available datasets before running our main computational pipeline. UniProt-SPROT and UniRef50 are our main sources of protein sequences. All sequences are cleaned only after an official, validated, automated download. In this context, 'prior processing' refers specifically to any data cleaning or manipulation done before the computational pipeline starts. In contrast, 'preprocessing' refers to steps such as tokenization and the removal of special characters from sequences, which occur immediately before the main pipeline and as part of it.

In the PPI experiment, we apply automated preprocessing to positive and negative ground truth links. Sequence cleaning is dynamic: when a pipeline component requests sequence data, the raw FASTA file is read and any character not representing one of the 20 standard amino acids (A, C, D, E, F, G, H, I, K, L, M, N, P, Q, R, S, T, V, W, Y) is filtered out. This standardizes the alphabet for all downstream models. Sequence lengths are capped at 10,000 characters to fit in memory and on the GPU. Tokenization, or preprocessing, occurs on the fly for all models tested, including our own. For models like Word2Vec or LSTMs, the cleaned sequence is used as a list of characters. For the ProtGram model, the cleaned sequence is split into overlapping n-grams.

For PPI ground truth, positive links are automatically downloaded from BioGrid, and negative links are obtained from Trabuco et al. (2012). Identifiers from these raw datasets are converted to canonical UniProtKB IDs using a mapping database built directly from the official UniProt ID mapping database. The standardized interaction pairs are saved in Parquet format and serve as the definitive ground truth for all evaluation tasks. Further preparation details are in the appendix.

3.2 Intrinsic Evaluation of *DirectGCN*

To establish the general graph representation learning capabilities of the *DirectGCN* architecture, we first evaluated it on commonly used public benchmark datasets for node classification. Though it is very important to note that

- **Datasets:** We selected standard GNN benchmark datasets: Cora, CiteSeer, PubMed, Cornell, Texas, and Wisconsin. For each, we evaluated performance on both their original (potentially directed) and explicitly symmetrized (undirected) versions.

Table 1. Statistics of standard datasets used in the benchmark evaluation.

Dataset	# Nodes	# Edges	# Features	# Classes
Cora	2,708	10,556	1,433	7
PubMed	19,717	88,648	500	3
Cornell	183	298	1,703	5
Texas	183	325	1,703	5
Wisconsin	251	515	1,703	5

- 279 • **Task & Setup:** The task was semi-supervised node classification using a fixed 10%/10%/80%
 280 train/validation/test split. All models were trained for 300 epochs using the Adam optimizer.
- 281 • **Baseline Models:** Graph Convolutional Network (GCN) [?], Graph Attention Network (GAT) Veličković
 282 et al. (2018), GraphSAGE Hamilton et al. (2018), Graph Isomorphism Network (GIN) Xu et al. (2019),
 283 and Tong et al.'s DiGCN Tong et al. (2020).
- 284 • **Results:** The goal of this evaluation was to validate *ProtGramDirectGCN* as a sound GNN archi-
 285 tecture. On high-homophily citation networks (Cora, CiteSeer, PubMed), *ProtGramDirectGCN*
 286 underperformed relative to simpler models like GCN and GAT. This is an expected outcome, as its
 287 complex, over-parameterized architecture is not well-suited for these tasks and struggles to converge
 288 effectively. However, on the heterophilic WebKB datasets (Cornell, Texas, Wisconsin), where rela-
 289 tionships are more complex, its performance was more competitive. This validates that the model is
 290 functional but highly specialized, justifying its application to our custom, non-homophilic n-gram
 291 graphs rather than general-purpose benchmarks. A summary of results is presented in Table 2.

Table 2 Model performance on directed datasets. Accuracy and F1-Score are reported as *mean ± std.* (M) denotes macro average. Bold indicates best performance.

Dataset	Model	Accuracy	F1-Score (M)	Precision (M)	Recall (M)
Cora	GCN	0.8722 ± 0.0088	0.8622 ± 0.0106	0.8629	0.8651
	GAT	0.8863 ± 0.0062	0.8754 ± 0.0099	0.8808	0.8723
	GIN	0.8671 ± 0.0103	0.8588 ± 0.0134	0.8637	0.8575
	ProtGramDirectGCN	0.8590 ± 0.0189	0.8480 ± 0.0256	0.8493	0.8497
	Tong et al.	0.8530 ± 0.0142	0.8407 ± 0.0172	0.8449	0.8400
PubMed	GCN	0.8631 ± 0.0047	0.8553 ± 0.0056	0.8573	0.8542
	GAT	0.8529 ± 0.0089	0.8456 ± 0.0103	0.8491	0.8445
	GIN	0.8716 ± 0.0052	0.8669 ± 0.0054	0.8652	0.8695
	ProtGramDirectGCN	0.8451 ± 0.0053	0.8370 ± 0.0067	0.8360	0.8392
	Tong et al.	0.8107 ± 0.0120	0.8000 ± 0.0116	0.8022	0.8007
Cornell	GCN	0.4101 ± 0.0608	0.2440 ± 0.0590	0.2406	0.2713
	GAT	0.4264 ± 0.0748	0.1684 ± 0.0432	0.1989	0.2202
	GIN	0.4862 ± 0.0770	0.3603 ± 0.0546	0.3682	0.3974
	ProtGramDirectGCN	0.5571 ± 0.0499	0.4104 ± 0.0837	0.5182	0.4061
	Tong et al.	0.5520 ± 0.0316	0.2976 ± 0.0547	0.3096	0.3356
Texas	GCN	0.3773 ± 0.0923	0.1640 ± 0.0403	0.1575	0.1770
	GAT	0.5464 ± 0.0567	0.2139 ± 0.0461	0.2163	0.2569
	GIN	0.4045 ± 0.0585	0.2199 ± 0.0369	0.2305	0.2414
	ProtGramDirectGCN	0.6940 ± 0.0202	0.5212 ± 0.0831	0.6044	0.5071
	Tong et al.	0.5353 ± 0.0540	0.2310 ± 0.0496	0.2390	0.2703
Wisconsin	GCN	0.4224 ± 0.0627	0.2491 ± 0.0702	0.2600	0.2641
	GAT	0.4898 ± 0.0937	0.2413 ± 0.0686	0.3067	0.2635
	GIN	0.4219 ± 0.0625	0.2876 ± 0.0757	0.2910	0.3000
	ProtGramDirectGCN	0.6293 ± 0.0423	0.3833 ± 0.0584	0.3835	0.4079

Continued on next page...

Table 2 – continued from previous page

Dataset	Model	Accuracy	F1-Score (M)	Precision (M)	Recall (M)
	Tong et al.	0.4975 ± 0.1026	0.2695 ± 0.0784	0.2934	0.2814

3.3 Construction of Hierarchical G_n Graphs

The primary dataset for our methodology is a hierarchy of global n-gram graphs, G_n , constructed from the UniProt Swiss-Prot sequence database.

- **Corpus:** We used the curated and reviewed UniProt Swiss-Prot dataset, containing 573,230 protein sequences.
- **Graph Construction:** For each n-gram level from $n = 1$ to $n = 4$, we constructed a separate graph G_n . The nodes V_n are the unique n-grams of length n found in the corpus. A directed edge (u, v) exists if n-gram v can be formed by shifting a one-character window over n-gram u . The edge weight w_{uv} is the total frequency of this transition across all sequences. This process resulted in four graphs of increasing size and complexity, as detailed in Table 3.

Table 3. Statistics of the constructed n-gram graphs (G_n).

n-gram Level (n)	# Nodes (Unique n-grams)	# Edges (Unique Transitions)
1	26	601
2	601	10,669
3	10,669	180,273
4	180,273	3,240,330

3.4 Learning N-gram Embeddings from G_n

The constructed G_n graphs serve as the foundation for learning informative vector representations (embeddings) for each n-gram. This is achieved through a self-supervised training task designed to force the model to understand the sequential grammar inherent in the protein sequences from which the graph was built.

- Self-Supervised Task: Next-Node Prediction:** For each n-gram node $u \in V_n$, we define its label y_u as its most likely successor in the sequence. This successor is determined by identifying the outgoing edge (u, v) with the highest transition frequency (weight) in the raw graph. The task for the GNN is therefore to predict this most probable next n-gram for every node in the graph. This objective compels the model to learn embeddings that encode the sequential and transitional logic of the n-gram language. An n-gram's representation becomes a function of not only its own identity but also the likely sequences it participates in.
- Hierarchical Training:** The training process is hierarchical. For the base level ($n = 1$), node features are randomly initialized. For each subsequent level $n > 1$, the initial features for a given n-gram node are generated by mean-pooling the final, learned embeddings of its two constituent (n-1)-gram nodes from the previously trained level. This creates a rich, multi-scale representation, where higher-order n-gram features are built upon the learned representations of their sub-components.
- Implementation Details:** The model for each level n is trained for a set number of epochs using the Adam optimizer and a negative log-likelihood loss function on the next-node prediction task. For larger graphs ($n \geq 3$), a Cluster-GCN Chiang et al. (2019) approach is used to partition the graph into mini-batches for memory-efficient training. The final output of this stage is a comprehensive set of learned embeddings for all n-grams at the highest level, $n = 4$.

3.5 Ablation Study to Optimize the Design of *ProtGram* – *DirectGCN*

Here we are going to understand the properties of the n-gram graph and its generated residue representation. This step is crucial as the smallest perturbation in the data or the model affect the final per protein embedding due to the hierarchical nature of the model.

3.5.1 The effect of the size of the data

The diversity of the protein sequences greatly affect the quality of the generated representations to understand this after we build the n-gram graphs we run our model alongside other models on a community detection task. The goal is to assess the different sizes of the generated n-gram graphs built from different sizes of sampled sequences.

333 3.6 Protein-Protein Interaction (PPI) Link Prediction

334 The ultimate test of the learned n-gram embeddings is their utility in a practical, downstream biological
335 task. We apply them to predict protein-protein interactions, a canonical link prediction problem.

- 336 • **Protein-Level Embedding Generation:** A single, fixed-size feature vector is generated for each
337 protein in the UniProt dataset. This is achieved by taking the sequence of each protein, identifying all
338 of its constituent 4-grams, retrieving their learned embeddings from the final *ProtGramDirectGCN*
339 model, and aggregating these vectors via mean pooling. This results in a single vector that summarizes
340 the global n-gram statistics for each protein. To standardize the feature space for comparison with other
341 methods, Principal Component Analysis (PCA) is applied to reduce the final embedding dimension to
342 64.
- 343 • **PPI Datasets:** A benchmark PPI dataset was compiled using known positive interactions from the
344 BioGRID database Oughtred et al. (2021) and high-quality negative interactions (non-interacting pairs)
345 from the experimentally-derived Russell Lab datasets Trabuco et al. (2012). This ensures a robust and
346 biologically relevant evaluation set.
- 347 • **Link Prediction Model:** A standard Multi-Layer Perceptron (MLP) was used as the classifier. For a
348 pair of proteins (P_a, P_b), the input to the MLP was the concatenation of their embedding vectors.
- 349 • **Evaluation and Baselines:** The model's performance is rigorously assessed using a 5-fold stratified
350 cross-validation scheme to ensure that results are robust and not dependent on a single random data
351 split. We measure performance using a suite of standard binary classification and ranking metrics,
352 including Area Under the ROC Curve (AUC), F1-Score, Precision, Recall. To contextualize our results,
353 we compare the performance of our *ProtGramDirectGCN*-derived embeddings against ProtT5
354 Elnaggar et al. (2022) available via UniProt. The exact same MLP architecture and evaluation protocol
355 are used for all embedding types to ensure a fair comparison. See Table 4, Figure 3,4,??.

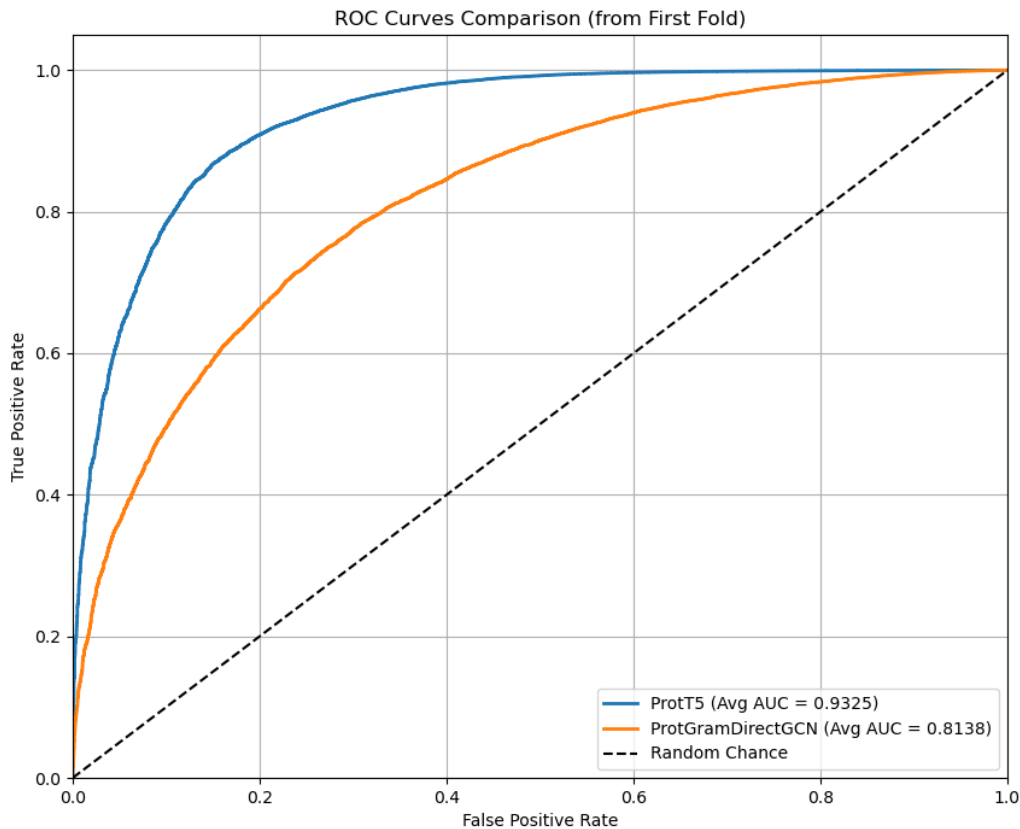


Figure 3. The plot displays the Receiver Operating Characteristic (ROC) curves comparing the performance of protein embeddings generated by the proposed *ProtGramDirectGCN* method against the state-of-the-art ProtT5 language model. The evaluation is for the downstream task of Protein-Protein Interaction (PPI) link prediction, with this specific chart illustrating the results from the first fold of a 5-fold cross-validation. The Area Under the Curve (AUC) for ProtT5 is 0.9325, indicating superior performance. The ProtGramDirectGCN-derived embeddings achieve a robust AUC of 0.8138. Both models perform significantly better than random chance (dashed line). This visualization confirms that while the proposed graph-based method captures a strong predictive signal for protein interactions, the ProtT5 model serves as a higher-performing benchmark in this experiment.

Table 4. Performance Comparison of Protein Embeddings on PPI Link Prediction (Averaged over 5 Folds)

Embedding Method	AUC	F1	Precision	Recall
ProtT5	0.9325±0.0005	0.9490±0.0021	0.9957	0.9065
ProtGramDirectGCN	0.8138±0.0052	0.8481±0.0074	0.9927	0.7403

4 DISCUSSION

356 This study introduced and evaluated a novel *ProtGramDirectGCN* model for learning representations
357 from a globally constructed, directed, dense, and weighted graph of amino acid residues G_n derived from
358 the UniProt dataset. The primary objective was to assess the efficacy of this approach for generating

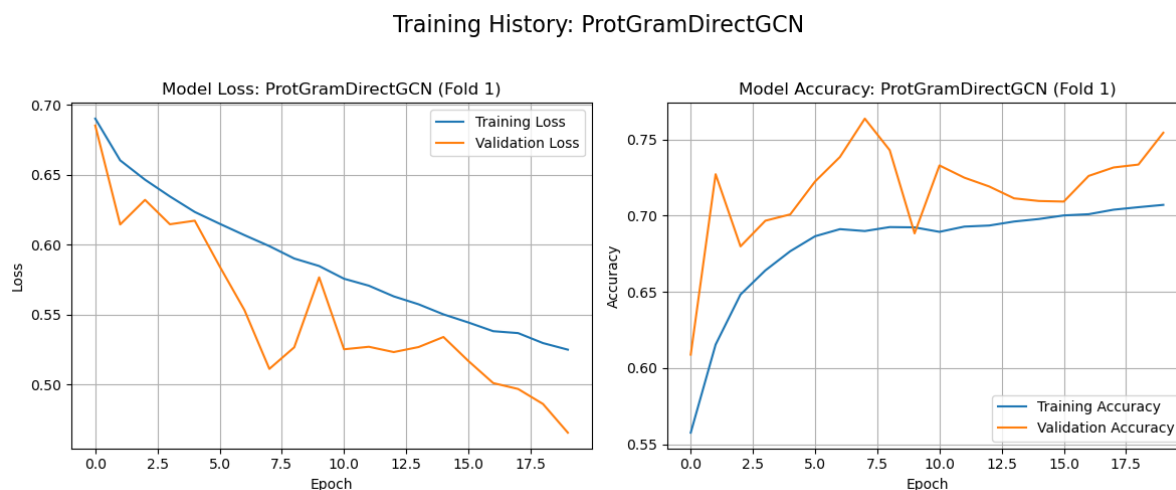


Figure 4. The figure illustrates the model's learning progress during the self-supervised, next-node prediction task on an n -gram graph G_n , representing the results from the first fold of a cross-validation run. Left Panel (Model Loss): This plot tracks the training loss (blue) and validation loss (orange) over 20 epochs. The steady decrease in training loss indicates the model is effectively learning from the data. The validation loss, while more volatile, generally follows the downward trend, suggesting the model is generalizing to unseen data within the graph. Right Panel (Model Accuracy): This plot shows the corresponding training and validation accuracy. Both metrics trend upward, demonstrating that the model's performance on the next-node prediction task improves as it is trained. Together, these plots confirm that the *ProtGramDirectGCN* model was successfully trained, effectively learning the underlying sequential patterns within the n -gram graph to generate meaningful embeddings for the downstream PPI task.

informative protein embeddings applicable to downstream biological prediction tasks, particularly Protein-Protein Interaction (PPI) link prediction. This section discusses the main findings, their implications, the limitations of the current work, and promising avenues for future research.

4.1 Summary of Findings

Our experimental evaluations spanned several stages: validating the core *ProtGramDirectGCN* architecture on standard GNN benchmarks and applying the derived protein-level embeddings to predict PPIs, including a comparison against state-of-the-art Protein Language Model (ProtT5) embeddings.

- **Intrinsic Evaluation of *DirectGCN*:** The benchmark results in Table 2 confirm that *ProtGramDirectGCN* is a functionally sound GNN. Its underperformance on high-homophily citation networks and competitive performance on more complex, heterophilic graphs highlights its specialization. The architecture is not designed as a general-purpose GCN but as a specialized tool for capturing the complex, directed, and weighted relationships present in our n -gram residue graphs.
- **Embedding G_n :** The hierarchical construction of n -gram graphs up to $n = 4$ (Table 3) resulted in a large, complex graph structure. The successful training of our model on this graph hierarchy demonstrates the feasibility of the approach. The key outcome of this stage is the set of high-dimensional embeddings for 180,273 unique 4-grams, which serve as the basis for our protein-level representations.
- **Efficacy for protein interaction prediction:** The performance on the downstream PPI link prediction task (Table ??) is the ultimate measure of our method's utility. The results show that the *ProtGramDirectGCN* was able to learn structural features from the protein sequences and hence was able to demonstrate excellent predictive power on the task with an AUC value above 80%. The

F1 score also demonstrates the model's precision even with a lowered recall and hence missing more positive samples due to the limited capacity of the model. This highly suggested that the construction and inference of the underlying directed graph of amino acid transitions in a hierarchical fashion captures structural and relational features across multiple proteins.

- **Comparison with ProtT5 Embeddings:** The comparison with ProtT5 embedding which are generated by the very powerful high capacity T5 encoder-decoder transformer model that is trained on the more comprehensive UniRef50 dataset is not meant as a head to head comparison but rather as a demonstration that hefty transformer architectures for specialized tasks like PPI can be contended with models that capture the underlying dynamics without having to rely on long context windows. The long range dependancies captured by ProtT5 are the reason why it is an efficient feature extractor for proteins. Yet those same dynamics can be captured from a lower level faint signal such as the simple transition directed graph of amino acids without long context windows. Both ProtT5 and *ProtGramDirectGCN* rely on computationally expensive preprocessing yet with *ProtGramDirectGCN* the significant decrease in the cost of model training especially when the technique gets more established will result in a paradigm shift when it comes to how we think about specialized tasks for LLMs in general.

4.2 Biological Significance

Protein-protein interaction (PPI) prediction is a bedrock in drug development, understanding drug efficacy, and many other crucial biomedical fields. *In-silico*, data-driven machine learning-based methods play an indispensable role in guiding *in-vitro* experiments, helping to create a refined pool of possible interactions ready for lab confirmation. Hence, enhancing the predictive power of *in-silico* methods directly contributes to increasing the efficiency of the entire research pipeline. While recent advancements such as AlphaFold have revolutionized structure prediction, offering indirect avenues for inferring interactions when combined with primary structure information, the most common approach in predicting protein interactions *in-silico* has heavily relied on advancements in language models or, more generally, foundation models. These typically employ techniques such as large language modeling and the Transformer architecture, with various configurations of layers, context windows, tokenization granularities, and pooling strategies to achieve per-protein embeddings capable of predicting links via similarity searches in downstream tasks.

Our research contributes to this landscape with several key points of significance:

- This work's primary novelty lies in constructing and utilizing a global, directed, dense, and weighted graph of amino acid residues G_n as a foundational source for feature learning. We explicitly model global sequence statistics and transition patterns by applying *ProtGramDirectGCN* to G_n . That offers an alternative and potentially complementary approach to methods primarily focusing on individual protein sequences (like PLMs) or direct GNN application to PPI network topology. In PLMs, the context window size often limits how far the residue might affect the overall embedding of the amino acid, since not all contexts in which the residue appeared or was linked to another functional group have been captured. Our approach ensures that that limitation is addressed at a more granular level, information was captured even with the limitations of G_n . In addition, the PPI network topology does not carry information from the structure of the proteins, presenting yet another critical feature that was not considered when applying a GNN.
- *ProtGramDirectGCN*, with its unique adjacency matrix preprocessing and its dual-path, hierarchical layer featuring distinct weights for incoming, outgoing, and shared information, combined via learnable coefficients, is specifically tailored to harness the rich, directed information within G_n .

- The improved predictive performance for PPIs. The protein embeddings derived from our *ProtGramDirectGCN* pipeline demonstrated competitive performance with state-of-the-art methods like ProtT5 for PPI link prediction. This suggests that capturing global n-gram co-occurrence and transition patterns provides a powerful signal for predicting interactions.
- Enhancing PPI prediction accuracy and interpretability has broad implications. By successfully training GNNs on G_n , our work suggests that meaningful biological signals related to residue function and context can be extracted from global sequence statistics when structured as a graph. The hierarchical structure (residues to proteins) and the potential for future incorporation of interpretability techniques (e.g., attention analysis, gradient-based methods) aim to move beyond "black box" predictions. That can help define lead residues or sequence motifs contributing to specific interactions, thereby guiding hypothesis generation for experimental testing and accelerating the functional annotation of uncharacterized proteins in expanding proteomic datasets. This work can enhance understanding of complex biological networks and cellular machinery by providing more precise interaction maps.
- Furthermore, identifying prominent interactions, particularly those responsible for disease mechanisms, lies at the core of new drug target finding and validation. Improved *in-silico* predictions, as pursued in this study, have direct implications for accelerating therapeutic development and advancing personalized medicine.
- The rigorous experimental design, including K-fold cross-validation for GNN evaluation on G_n , careful selection of positive and negative samples for PPI link prediction, and strategies to mitigate data leakage, contributes to a highly validated and robust assessment of our proposed methods.
- Future combination of our learned representations with advanced visualization programs, spurred on by findings from embedding analyses, may also aid scientists in interpreting high-complexity interaction networks. Lastly, this study contributes to the noble endeavor of untangling the complex molecular interactions that underpin life.

4.3 Limitations

While this study presents promising results, several limitations should be acknowledged:

- Due to computational constraints, G_n was constructed based on UniProt Swiss-Prot. While providing a high-quality reviewed set, more comprehensive datasets, such as UniRef50/90 or the full UniProtKB, could enrich G_n at a significant computational cost. Initial features for 1-gram nodes in G_1 were initialized to identity. Including physicochemical properties as initial features could enhance learning and interpretability. Simple attention pooling was used to generate protein-level embeddings. More advanced pooling mechanisms were not exhaustively explored and might yield improved representations. Evaluation centered on PPI link prediction. The utility of embeddings for additional tasks remains to be explored.
- While the design of *ProtGram – DirectGCN* is detailed, and we provided some insight via attention map visualization of the role of distinct n-grams and how they map to functional groups, direct interpretation of what specific n-gram relationships contribute most to its performance or downstream PPI predictions currently relies on indirect evaluation through task performance. Deeper interpretability studies are warranted.

- The PPI link prediction task relies on the Russell Lab negative dataset Trabuco et al. (2012), which, while experimentally grounded, has inherent assumptions and potential biases based on Yeast two-Hybrid limitations. The choice of negative samples can significantly impact the reported performance of PPI prediction.
- The current framework primarily relies on sequence-derived information for constructing G_n and generating protein embeddings. Direct integration of 3D structural information was not part of this specific study, but it is a key area for future enhancement, as our focus was on leveraging the limited available information and training data.

4.4 Future Work

The findings and limitations of this study open several avenues for future research. Future iterations will explore constructing G_n with richer edge definitions. For example, we could move beyond simple findings and limitations of this study open several avenues for future research. First, future iterations will explore constructing G_n with richer edge definitions. For example, we could move beyond simple transitions to incorporate longer-range co-occurrences, apply weights from substitution matrices, or use more informative initial node features for amino acids. Second, *ProtGram – DirectGCN* could be extended by incorporating attention mechanisms within its directional layers. In addition, we could explore more complex architectures with advanced normalization schemes. Another focus will be on introducing and crafting training tasks that are more specific to proteins and their interactions. We also plan to adapt the model to other tasks, such as predicting Gene Ontology labels for individual proteins. Furthermore, integrating 3D structural information is a promising direction for future research. This work focuses solely on the limitations of training data and information, which are represented by relying on a limited subset of protein sequences. To address this, we could use predicted contact maps to inform the edges in G_n or in peptide-level graphs similar to Sledzieski et al. (2021), or incorporate residue-level structural features into the initial residue embeddings. Expanding the framework to explicitly model the hierarchical nature of protein organization (residues $\rightarrow$ peptides $\rightarrow$ proteins $\rightarrow$ interactions $\rightarrow$ interaction networks) and exploring second-degree graphs (graphs of interactions) presents a compelling research direction. Applying the learned protein representations to a broader array of prediction tasks beyond PPIs will be crucial for generating biological insights. Lastly, investing in advanced interpretability techniques will help understand the “black box,” and further optimizing the construction of G_n and training *ProtGram – DirectGCN* for even larger sequence datasets will maximize the information captured.

5 CONCLUSION

This paper introduces a novel approach for protein representation learning, which has been shown to enable *in-silico* protein-protein interaction prediction via a simpler yet expressive learning model. The method focuses on a novel data model that infers hierarchical global n-gram graphs from protein sequences. In these graphs, n-grams—defined as contiguous sequences of n amino acids in proteins—form the nodes, and edges represent relationships between these n-gram sequences. A custom directed graph convolution learning model, *ProtGram – DirectGCN*, is introduced. This model is designed to learn from protein graphs with directed edges (edges have direction, indicating the flow from one n-gram to another), heterophily (connections often occur between nodes representing different types of n-grams), and weighted edges (edges have numerical weights that may represent the strength or frequency of the relationship). The model learned distinctive features that capture protein relations, even with limited training data. This offers a valuable and computationally distinct alternative to large-scale Protein Language Models (PLMs), such

as ProtT5, under the evaluated conditions. Developing specialized Graph Neural Network architectures, such as *ProtGram – DirectGCN*, can harness global statistical patterns from entire sequence databases, thereby reducing reliance on computationally expensive PLMs. Our approach demonstrates promise for computational protein biology. In the future, graph-based representations will be enriched with multi-modal data, including explicit structural information. The scope of application will expand to more biological problems. Ultimately, this work aims to provide a deeper understanding of the molecular interactions that govern life by introducing new methods for analyzing and understanding protein and gene interactions.

CONFLICT OF INTEREST STATEMENT

The authors declare that the research was conducted without commercial or financial relationships that could create a conflict of interest.

AUTHOR CONTRIBUTIONS

IAE: developed the preprocessing of protein sequences in addition to designing the directed GCN architecture HT: optimized the performance of the GCN model via parallelization, in addition to describing in writing the article PG: performed the evaluation of the model and contributed to the writing of the article

FUNDING

No funding was involved in this work.

SUPPLEMENTAL DATA

Please see supplementary material for more information.

DATA AVAILABILITY STATEMENT

The datasets and codes for this study can be found at <https://github.com/iebeid/ProtGram-DirectGCN/tree/v2>.

REFERENCES

- Anfinsen, C. B. (1973). Principles that govern the folding of protein chains. *Science* 181, 223–230. doi:10.1126/science.181.4096.223. Publisher: American Association for the Advancement of Science
- Chiang, W.-L., Liu, X., Si, S., et al. (2019). Cluster-GCN: An Efficient Algorithm for Training Deep and Large Graph Convolutional Networks. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining* (Association for Computing Machinery), Kdd '19, 257–266. doi:10.1145/3292500.3330925
- Cho, K., van Merriënboer, B., Gulcehre, C., et al. (2014). Learning Phrase Representations using RNN Encoder–Decoder for Statistical Machine Translation. In *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, eds. A. Moschitti, B. Pang, and W. Daelemans (Association for Computational Linguistics), 1724–1734. doi:10.3115/v1/D14-1179
- Dill, K. A. and MacCallum, J. L. (2012). The protein-folding problem, 50 years on. *science* 338, 1042–1046. doi:10.1126/science.1219021. Publisher: American Association for the Advancement of Science

- Elnaggar, A., Heinzinger, M., Dallago, C., et al. (2022). ProtTrans: Toward Understanding the Language of Life Through Self-Supervised Learning. *IEEE Trans Pattern Anal Mach Intell* 44, 7112–7127. doi:10.1109/tpami.2021.3095381
- [Dataset] Hamilton, W. L., Ying, R., and Leskovec, J. (2018). Inductive Representation Learning on Large Graphs. doi:10.48550/arXiv.1706.02216. ArXiv:1706.02216 [cs]
- Hochreiter, S. and Schmidhuber, J. (1997). Long Short-Term Memory. *Neural Comput.* 9, 1735–1780. doi:10.1162/neco.1997.9.8.1735. 655 citations (INSPIRE 2025/8/30) 654 citations w/o self (INSPIRE 2025/8/30)
- Jeh, G. and Widom, J. (2002). SimRank: a measure of structural-context similarity. In *Proceedings of the eighth ACM SIGKDD international conference on Knowledge discovery and data mining* (Association for Computing Machinery), Kdd '02, 538–543. doi:10.1145/775047.775126
- Jha, K., Saha, S., and Singh, H. (2022). Prediction of protein–protein interaction using graph neural networks. *Sci Rep* 12, 8360. doi:10.1038/s41598-022-12201-9. Publisher: Nature Publishing Group
- Jumper, J., Evans, R., Pritzel, A., et al. (2021). Highly accurate protein structure prediction with AlphaFold. *Nature* 596, 583–589. doi:10.1038/s41586-021-03819-2. Publisher: Nature Publishing Group
- Kipf, T. N. and Welling, M. (2017). Semi-Supervised Classification with Graph Convolutional Networks doi:10.48550/arXiv.1609.02907. 152 citations (INSPIRE 2025/8/30) 152 citations w/o self (INSPIRE 2025/8/30) arXiv:1609.02907 [cs.LG]
- Ofer, D., Brandes, N., and Linial, M. (2021). The language of proteins: NLP, machine learning & protein sequences. *Computational and Structural Biotechnology Journal* 19, 1750–1758. doi:10.1016/j.csbj.2021.03.022
- Oughtred, R., Rust, J., Chang, C., et al. (2021). The BioGRID database: A comprehensive biomedical resource of curated protein, genetic, and chemical interactions. *Protein Science* 30, 187–200. doi:10.1002/pro.3978. Publisher: Wiley Online Library
- Perkins, J. R., Diboun, I., Dessailly, B. H., et al. (2010). Transient protein-protein interactions: structural, functional, and network properties. *Structure* 18, 1233–1243. doi:10.1016/j.str.2010.08.007. Publisher: Elsevier
- Rao, V. S., Srinivas, K., Sujini, G. N., et al. (2014). Protein-Protein Interaction Detection: Methods and Analysis. *Int J Proteomics* 2014, 147648. doi:10.1155/2014/147648
- Scannell, J. W., Blanckley, A., Boldon, H., et al. (2012). Diagnosing the decline in pharmaceutical R&D efficiency. *Nat Rev Drug Discov* 11, 191–200. doi:10.1038/nrd3681. Publisher: Nature Publishing Group
- Scarselli, F., Gori, M., Tsoi, A. C., et al. (2009). The Graph Neural Network Model. *IEEE Trans. Neural Networks* 20, 61–80. doi:10.1109/tnn.2008.2005605. Conference Name: IEEE Transactions on Neural Networks
- Scott, D. E., Bayly, A. R., Abell, C., et al. (2016). Small molecules, big targets: drug discovery faces the protein–protein interaction challenge. *Nat Rev Drug Discov* 15, 533–550. doi:10.1038/nrd.2016.29. Publisher: Nature Publishing Group
- Sledzieski, S., Singh, R., Cowen, L., et al. (2021). D-SCRIPT translates genome to phenome with sequence-based, structure-aware, genome-scale predictions of protein-protein interactions. *cels* 12, 969–982.e6. doi:10.1016/j.cels.2021.08.010. Publisher: Elsevier
- The UniProt Consortium (2023). UniProt: the Universal Protein Knowledgebase in 2023. *Nucleic Acids Research* 51, D523–d531. doi:10.1093/nar/gkac1052
- [Dataset] Tong, Z., Liang, Y., Sun, C., et al. (2020). Directed Graph Convolutional Network

- 574 Trabuco, L. G., Betts, M. J., and Russell, R. B. (2012). Negative protein–protein interaction datasets derived
575 from large-scale two-hybrid experiments. *Methods* 58, 343–348. doi:10.1016/j.ymeth.2012.07.028
- 576 Vaswani, A., Shazeer, N., Parmar, N., et al. (2017). Attention Is All You Need 388 citations (INSPIRE
577 2025/8/30) 388 citations w/o self (INSPIRE 2025/8/30) arXiv:1706.03762 [cs.CL]
- 578 Veličković, P., Cucurull, G., Casanova, A., et al. (2018). Graph Attention Networks doi:10.48550/
579 arXiv.1710.10903. 58 citations (INSPIRE 2025/8/30) 56 citations w/o self (INSPIRE 2025/8/30)
580 arXiv:1710.10903 [stat.ML]
- 581 Vidal, M., Cusick, M. E., and Barabási, A.-L. (2011). Interactome Networks and Human Disease. *Cell*
582 144, 986–998. doi:10.1016/j.cell.2011.02.016
- 583 Xu, K., Hu, W., Leskovec, J., et al. (2019). How Powerful are Graph Neural Networks? 33 citations
584 (INSPIRE 2025/8/30) 32 citations w/o self (INSPIRE 2025/8/30) arXiv:1810.00826 [cs.LG] version: 3
- 585 Zhou, H., Wang, W., Jin, J., et al. (2022). Graph Neural Network for Protein–Protein Interaction Prediction:
586 A Comparative Study. *Molecules* 27, 6135. doi:10.3390/molecules27186135. Number: 18 Publisher:
587 Multidisciplinary Digital Publishing Institute